

Function families

Topics: polynomial, exponential, logarithmic, and trigonometric functions.

Goal: learn the main types of functions, recognise their graphs, and compare how quickly they grow.

Today's question: **What can the shape of a function tell us before we calculate?**

Session	Date
4 of 15	Thu 10 Sep 2026

Where session 4 sits

1 · Course overview and algebra refresher

2 · Sets and set operations

3 · Functions

4 · Function families

5 · Theorem proving

6 · Mathematical induction

7 · Combinatorics and the binomial theorem

8 · Midterm

9 · GCD and divisibility

10 · Modular arithmetic

11 · Vectors

12 · Polynomials

13 · Complex numbers

14 · Course recap and exam preparation

15 · Final exam

TODAY

Polynomial, exponential/logarithmic, and trigonometric families.

Functions, in one page

ANATOMY	
$f : A \rightarrow B$	Domain, codomain, one output each
graph	In $A \times B$, each a once
fails	No output, or two
$f(A)$	Image: what is hit
codomain	Promised; the image is proved
$f^{-1}(S)$	Preimage: set to set, always
$g \circ f$	$g(f(a))$, f first. Sets match
order	$g \circ f \neq f \circ g$ usually

BEHAVIOUR	
Injective	$f(a_1) = f(a_2) \Rightarrow a_1 = a_2$: never twice
Surjective	Every b is hit
Bijective	Both: each b once
to prove	Assume equal images, or construct a
to disprove	One witness suffices
f^{-1}	Iff f bijects. Not $1/f$
restrict	Shrink until bijective; gives arcsin
$\cos \theta, \sin \theta$	On $x^2 + y^2 = 1$; period 2π

WARM-UP

Eventually larger: x^2 or 2^x ? Why isn't $x = 1$ enough?

One transformation language works for every family

DEFINITION

Start with the graph $y = f(x)$. For

$$g(x) = a f(b(x - h)) + k \quad (b > 0),$$

read from the **inside out**: change x first, then change y .

EXAMPLE

If $f(x) = x^2$, then $f(x - 2) = (x - 2)^2$
: the vertex moves from $(0, 0)$ to $(2, 0)$.

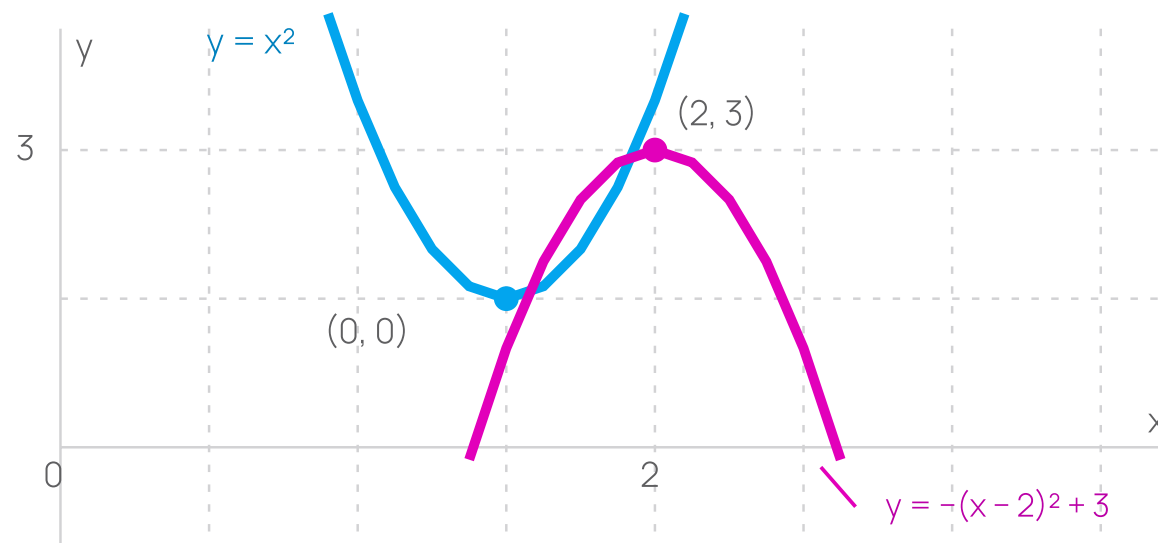
CHANGE	EFFECT ON THE GRAPH
h	move h units horizontally (right if $h > 0$)
k	move k units vertically (up if $k > 0$)
$ a $	make the graph $ a $ times taller; if $a < 0$, turn it upside down
b	make the graph $1/b$ times as wide

Track one familiar graph as it moves

EXAMPLE Start with $f(x) = x^2$. The graph of

$$g(x) = -(x - 2)^2 + 3$$

moves right 2, turns upside down, and moves up 3. The vertex moves from $(0, 0)$ to $(2, 3)$.



Polynomials

The formula tells us about the graph. Examples: 3 , $-2x + 5$, $x^2 - 4x + 3$, $2x^5 - x$.

In a simplified polynomial, the variable is not in a denominator, under a root, or in an exponent.

Question: Which part of a polynomial tells us what its graph does far left and far right?



Degree is the largest exponent with a coefficient that is not zero

DEFINITION A polynomial has the form

$$p(x) = a_n x^n + \cdots + a_1 x + a_0,$$

where $n \geq 0$ and $a_n \neq 0$. Its **degree** is n , and a_n is its **leading coefficient**. For this session, the coefficients a_i are real numbers.

WORKED

POLYNOMIAL	DEGREE	LEADING COEFFICIENT
$-4x^3 + 7x - 1$	3	-4
$5x^2 - 2x^5$	5	-2
6	0	6

METHOD Write the largest power first before you find the degree.

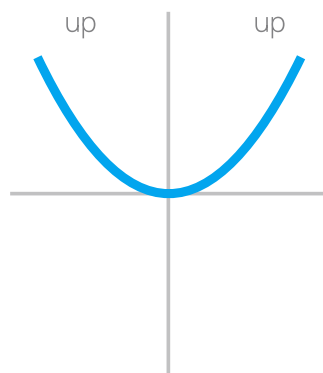
The highest-power term tells us what happens at the ends

DEGREE n	LEADING COEFFICIENT a_n	LEFT END	RIGHT END
even	positive	up	up
even	negative	down	down
odd	positive	down	up
odd	negative	up	down

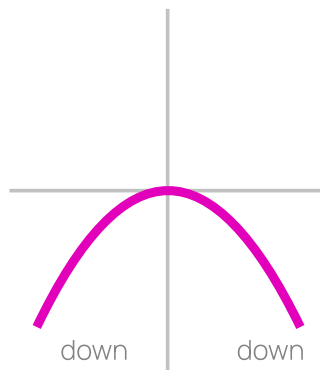
READING IT When x is very large positive or very large negative, $p(x)$ behaves like $a_n x^n$. Here **up** means the value becomes very large positive; **down** means it becomes very large negative.

EXAMPLE $p(x) = -2x^5 + 7x^2 - 1$ has odd degree and a negative leading coefficient, so it goes **up on the left** and **down on the right**.

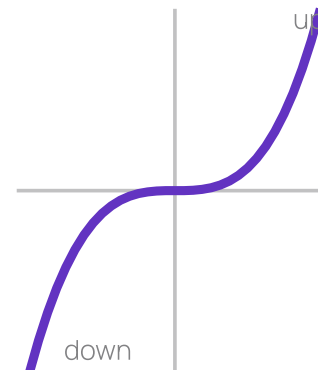
Four leading terms give the four possible end patterns



even, positive: x^2



even, negative: $-x^2$



odd, positive: x^3



odd, negative: $-x^3$

USE THIS WITH THE TABLE Ignore the middle of a polynomial graph here. The degree's parity and the leading coefficient determine only what happens far to the left and right.

Roots are where the graph meets the x -axis

DEFINITION

A number r is a **root** (or zero) of p when $p(r) = 0$.

FACTOR THEOREM

$$p(r) = 0 \iff (x - r) \text{ is a factor of } p(x).$$

EXAMPLE

$p(x) = x^3 - 4x = x(x^2 - 4) = x(x - 2)(x + 2)$,
so the roots are -2 , 0 , and 2 .

AT MOST

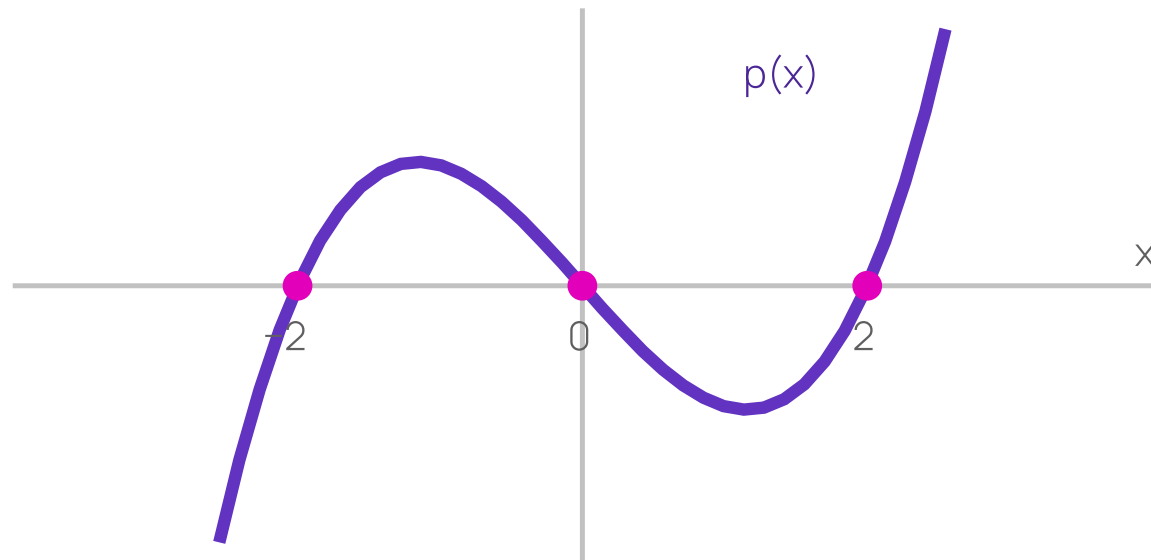
A nonzero polynomial of degree n has **at most** n different real roots.

For this cubic, each root is an x -axis crossing

READ THE CROSSINGS

$$p(x) = x^3 - 4x = x(x - 2)(x + 2)$$

The graph crosses the x -axis at $x = -2$, 0 , and 2 . Those x -coordinates are the roots.



Read a quadratic from its formula

ACTIVITY 1 GROUPS OF THREE

12 MIN

For $q(x) = -(x - 2)^2 + 3$:

1. State the degree and leading coefficient.
2. Describe the transformation from $y = x^2$.
3. State the vertex and end behaviour.
4. Find the roots in exact form (do not use decimals).

Work outwards from the completed square

METHOD

1. Expand mentally to get the degree; the number in front of the square is the leading coefficient.
2. Match q against $a f(b(x - h)) + k$ with $f(x) = x^2$: read h , then k , then a .
3. The vertex sits where the square is zero, so solve $x - h = 0$.
4. For roots, isolate the square and take **both** square roots.

ANSWER Degree 2, leading coefficient -1 ; shift right 2, up 3, then reflect vertically; vertex $(2, 3)$; both ends down.

$$-(x - 2)^2 + 3 = 0 \iff (x - 2)^2 = 3$$

so $x - 2 = \pm\sqrt{3}$ and $x = 2 \pm \sqrt{3}$.

THE TRAP Answering $\pm\sqrt{3}$ drops the horizontal shift. Undo the square first, then undo the shift.

Exponentials and logarithms

Repeated multiplication and its inverse. If adding 1 to the input always multiplies the output by the same number, the pattern is exponential.

$$2^0 = 1, \quad 2^1 = 2, \quad 2^2 = 4, \quad 2^3 = 8$$

A logarithm answers the reverse question: $2^x = 8$ asks: "What exponent gives 8?"



An exponential function has the variable in the exponent

DEFINITION

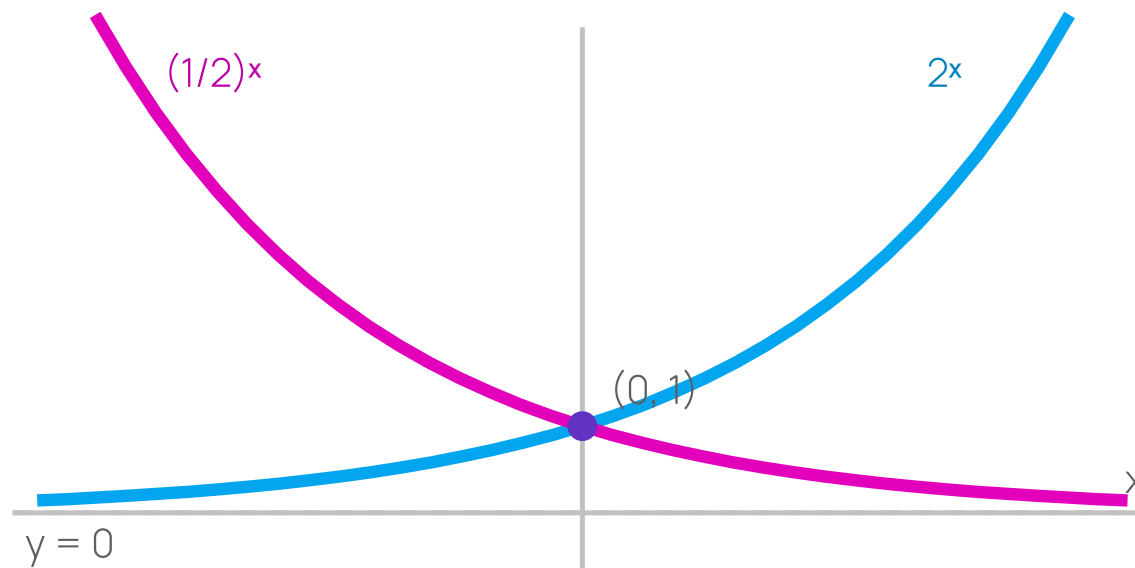
$$f(x) = b^x, \quad b > 0, b \neq 1$$

Every exponential graph passes through $(0, 1)$ because $b^0 = 1$. The line $y = 0$ is a horizontal **asymptote**: the graph approaches it but never reaches it.

BASE	BEHAVIOUR	DOMAIN AND RANGE
$b > 1$	growth	domain $\mathbb{R}$, range $(0, \infty)$
$0 < b < 1$	decay	domain $\mathbb{R}$, range $(0, \infty)$

Growth and decay share a point, then move in opposite directions

COMPARE Both $y = 2^x$ and $y = (1/2)^x$ pass through $(0, 1)$. For every real x , both stay above the x -axis. The dashed line is the asymptote $y = 0$.



An exponential model uses the same multiplier at every step

MODEL

$$N(t) = N_0 r^t,$$

where $N_0 > 0$ is the starting amount and $r > 0$ is the multiplier per time unit.

MULTIPLIER r	INTERPRETATION
$r = 1.20$	increase by 20% per time unit
$r = 0.85$	decrease by 15% per time unit

EXAMPLE $N(t) = 100(1.2)^t$ gives $N(3) = 172.8$. The time unit must be named: hours, days, years, or something else.

Make the bases match, then set the exponents equal

RULE For $b > 0$ and $b \neq 1$,

$$b^u = b^v \implies u = v.$$

WORKED EXAMPLE

$$3^{2x-1} = 27$$

$$3^{2x-1} = 3^3$$

$$2x - 1 = 3$$

$$x = 2$$

Each line has the same solution as the line before it.

A logarithm is an exponent

DEFINITION For $b > 0$, $b \neq 1$, and $x > 0$,

$$\log_b x = y \iff b^y = x.$$

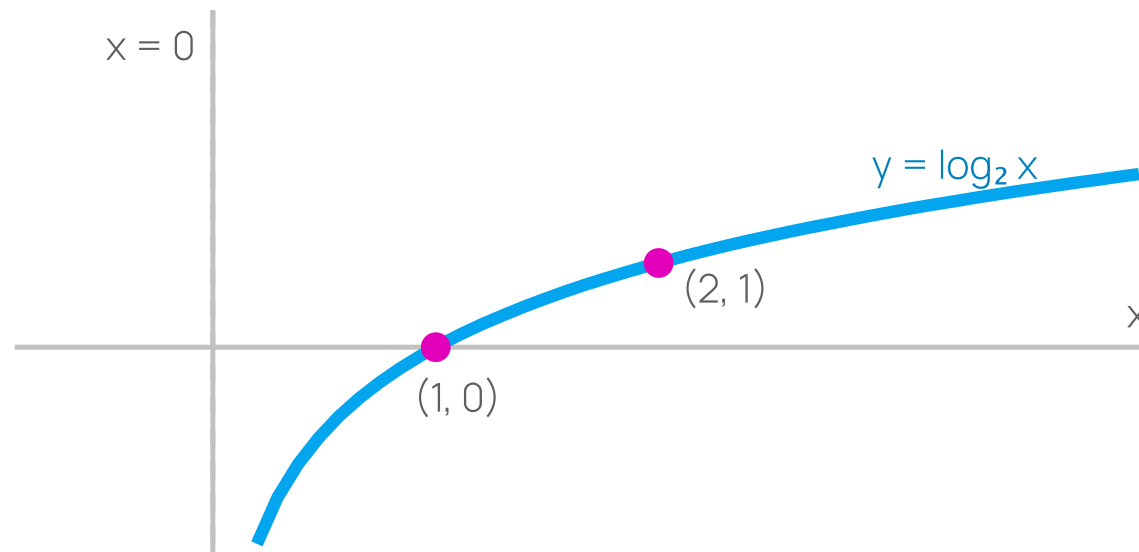
EXAMPLES

$$\log_2 8 = 3, \quad \log_{10}(0.01) = -2, \quad \ln(e^5) = 5.$$

DOMAIN $y = \log_b x$ has domain $(0, \infty)$ and range $\mathbb{R}$. There is no real value for $\log_b 0$ or $\log_b(-4)$.

The logarithm graph explains its domain

READ THE GRAPH $y = \log_2 x$ crosses at $(1, 0)$ and passes through $(2, 1)$. It exists only to the right of the y -axis: x must be positive.



Log laws change multiplication into addition

LAWS For $u > 0$ and $v > 0$:

$$\log_b(uv) = \log_b u + \log_b v$$

$$\log_b\left(\frac{u}{v}\right) = \log_b u - \log_b v$$

$$\log_b(u^c) = c \log_b u$$

THIS IS USUALLY FALSE

$$\log_b(u + v) \neq \log_b u + \log_b v \quad \text{in general.}$$

Counterexample: with $u = v = 1$, the two sides are $\log_2(1 + 1) = 1$ and $\log_2 1 + \log_2 1 = 0$.

First find the values of x that are allowed

ACTIVITY 2 PAIRS

14 MIN

Solve over the real numbers: $\log_2(x - 1) + \log_2(x + 1) = 3$.

1. Find the values of x allowed by both logarithms.
2. Combine the logarithms using a log law.
3. Solve the resulting equation.
4. Check every possible solution in the original equation.

Domain first, then algebra, then check

METHOD

1. Write the domain before touching the equation: every $\log_2$ input must be positive.
2. Combine the two logarithms with the product law, which needs both inputs positive.
3. Undo the logarithm using $\log_2 A = 3 \iff A = 2^3$.
4. Test every candidate against step 1, then substitute into the original equation.

ANSWER $x - 1 > 0$ and $x + 1 > 0$ give $x > 1$. Then

$\log_2((x - 1)(x + 1)) = 3$, so $x^2 - 1 = 8$, $x^2 = 9$, $x = \pm 3$. Only $x = 3$ lies in the domain. Check: $\log_2 2 + \log_2 4 = 1 + 2 = 3$.

THE TRAP $x = -3$ solves $x^2 = 9$ but not the original equation – squaring is where extra solutions appear.

Change a log base; compare growth only for large n

BASE CHANGE When $x > 0$, $a > 0$, $b > 0$, and neither base is 1:

$$\log_b x = \frac{\log_a x}{\log_a b}.$$

So changing the log base only multiplies the answer by a constant.

GROWTH For fixed $d > 1$, $k > 0$, and $c > 1$, when the positive integer n becomes large,

$$\log_d n < n^k < c^n$$

This is called an **eventual** comparison: the inequality is true once n is large enough, but it may not be true for every $n \geq 1$.

Rank the functions when n is large

ACTIVITY 3 GROUPS OF THREE

12 MIN

Put these in order from slowest growth to fastest growth when n is large: $\log_2 n$, n , $n \log_2 n$, n^2 , 2^n .

1. Why is $n \log_2 n$ larger than n once n is large?
2. Why does one numerical example not prove the full ranking?
3. Which expression is most likely to grow too quickly in an algorithm?

Compare neighbours, not all five at once

METHOD

1. Commit to a candidate order first; then you only have four adjacent pairs to justify.
2. For each pair, divide one by the other and ask what happens as n grows.
3. Name the n from which your claim holds — that is what "eventually" means.

ANSWER

$$\log_2 n < n < n \log_2 n < n^2 < 2^n$$

For $n > 2$, $\log_2 n > 1$, so $n \log_2 n > n$. And $n \log_2 n < n^2$ because $\log_2 n < n$. The exponential is the dangerous one for large inputs.

THE TRAP

At $n = 2$, $n^2 = 4 = 2^n$. One value can tie or even reverse the order, so it proves nothing.

Trigonometric functions

Graphs that repeat. From the unit circle, $-1 \leq \sin x \leq 1$ and $-1 \leq \cos x \leq 1$ for $x \in \mathbb{R}$;

and, in radians, $\sin(x + 2\pi) = \sin x$ and $\cos(x + 2\pi) = \cos x$.

Repeating graphs can model cycles: daylight, tides, sound, rotation, and seasonal demand.



Read a sine model from its parameters

DEFINITION

$$y = A \sin(B(x - C)) + D, \quad A \neq 0, B > 0$$

If $A < 0$, the wave is turned upside down before it is moved by D .

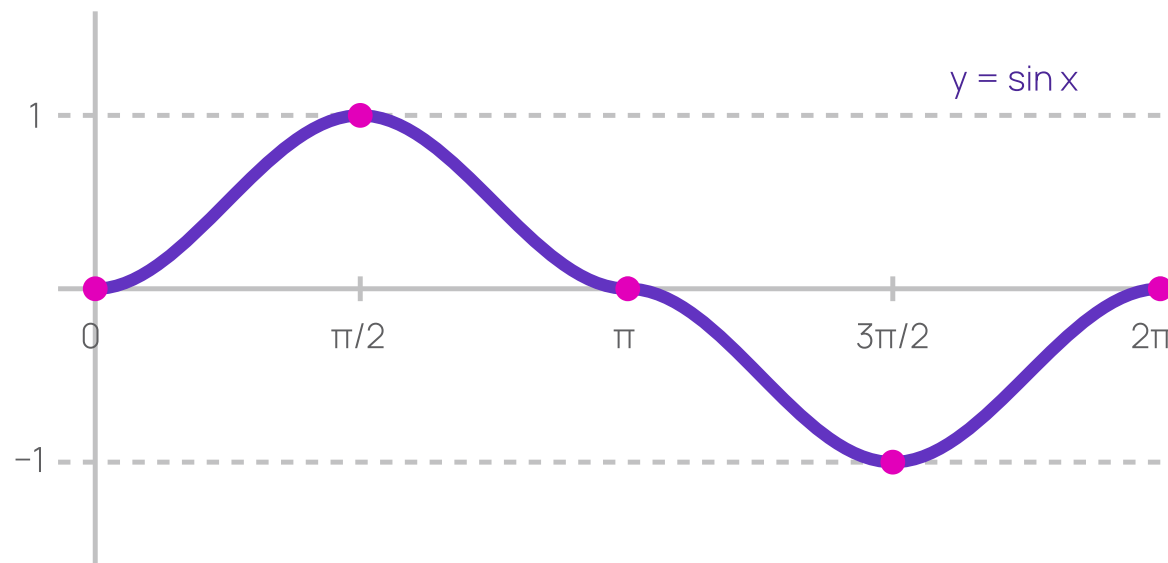
THE PARAMETERS

FEATURE	VALUE
midline	$y = D$
amplitude (distance from the midline to a peak)	$ A $
period	$\frac{2\pi}{B}$
horizontal shift	move C units horizontally (right if $C > 0$)
range	$[D - A , D + A]$

One sine cycle gives five useful landmarks

QUARTER-PERIOD STEPS For $y = \sin x$, start at $(0, 0)$

. Moving one quarter period at a time gives a peak, a midline crossing, a trough, and the next crossing. The full cycle has length 2π .



Worked model: daily temperature

MODEL Let t be hours after midnight. Consider

$$T(t) = 4 \sin\left(\frac{\pi}{12}(t - 3)\right) + 28.$$

The maximum occurs at $t = 9$ hours; the minimum occurs at $t = 21$ hours.

FEATURE	INTERPRETATION
amplitude 4	temperature varies 4° from its average
midline 28	average temperature is 28°
period 24	one complete cycle takes 24 hours
range [24, 32]	lowest and highest temperatures predicted by the model

Interpret before calculating

ACTIVITY 4 PAIRS

12 MIN

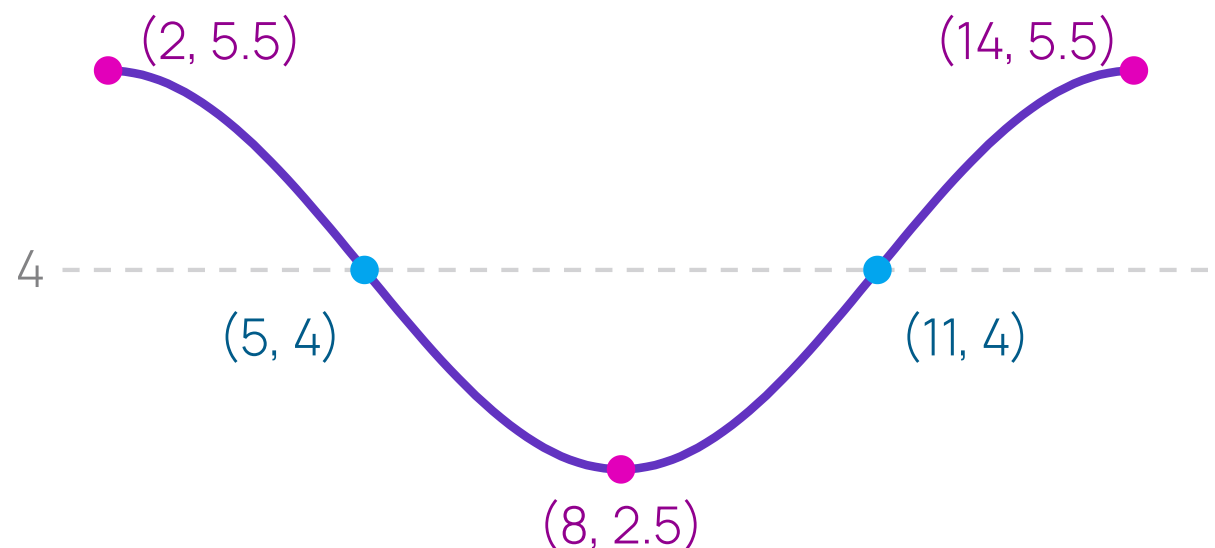
For $H(t) = 1.5 \cos\left(\frac{\pi}{6}(t - 2)\right) + 4$:

1. Find the amplitude, midline, period, and range.
2. When does the first maximum after $t = 0$ occur?
3. Sketch one cycle. Label a maximum, a minimum, and two points where the graph crosses its midline.

Read the parameters, then walk one cycle in quarter steps

METHOD

1. Match $A \cos(B(t - C)) + D$: here $A = 1.5$, $B = \frac{\pi}{6}$, $C = 2$, $D = 4$.
2. Amplitude $|A| = 1.5$, midline $y = 4$, period $\frac{2\pi}{B} = 12$, range $[2.5, 5.5]$.
3. Cosine peaks when its input is 0, so $t - 2 = 0$ gives $t = 2$, then every 12.
4. Step a quarter period (3) at a time to land the crossings and the minimum.



THE TRAP The period is **12**, not $\frac{\pi}{6}$: B sits *inside* the cosine, so divide 2π by it.

Work independently

1 For $p(x) = x^3 - 4x$, describe what happens on the far left and far right, and find all real roots.

2 Solve $5^{2x} = 125$.

3 For $y = 3 \cos\left(\frac{\pi}{4}(t + 2)\right) - 1$, state the amplitude, period, horizontal shift, and range.

RULE Write one sentence of justification for each answer. A correct number without its reason is incomplete.

What today was about

- The **leading term** tells us what a polynomial graph does at the ends; roots are where it meets the x -axis.
- An exponential has a **constant multiplier**. A logarithm records the exponent, and its input must be positive.
- For large inputs: **logarithmic** < **polynomial** < **exponential** growth.
- A trigonometric model describes repetition through **amplitude**, **period**, **shift**, and **midline**.

Problem Set 4

Growth comparisons, legal logarithm manipulation, and one periodic-model interpretation with written justification.

Next session: **theorem proving** — direct proof, contrapositive, contradiction, and counterexample.